

RUPESH SEN

Kolkata, India

☎ +91-9883473328

✉ rsrupeshsen@gmail.com

🌐 [LinkedIn](#)

🐙 [GitHub](#)

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Summary

Computer Science undergraduate at Manipal Institute of Technology with strong fundamentals in Python, Java, and full-stack development. Experienced in ML model development, ETL pipelines, data preprocessing, and test automation. Built AI-powered products integrating generative AI, RAG techniques, and computer vision. Certified in IBM Generative AI. Eager to take end-to-end ownership of AI use cases with minimal supervision in fast-paced environments.

Technical Skills

Programming: Python, Java, JavaScript, TypeScript, SQL, C

AI / ML: RAG, Prompt Engineering, Generative AI, OpenAI API, TensorFlow, PyTorch, NLP, Predictive Modeling, Supervised/Unsupervised Learning

Testing & QA: Manual Testing, Unit/Integration Testing, API Testing, TDD, JUnit, PyTest, Selenium

Data Engineering: ETL Pipelines, Data Preprocessing, MySQL, PostgreSQL, MongoDB, Redis

Backend / Frontend: Django, Flask, Spring Boot, Node.js, React, React Native, Tailwind CSS, REST APIs

DevOps / Cloud: Docker, Kubernetes, Jenkins, AWS, GCP, Terraform, Git, GitHub Actions, CI/CD

Education

Manipal Institute of Technology

2022 – 2026

B.Tech in Computer Science & Engineering (Minor: Computational Intelligence)

Manipal, Karnataka

Projects

CodeFolio | *React, TypeScript, Tailwind CSS, Vite, OAuth, REST APIs* | [Live Demo](#) | [GitHub](#)

- Built a full-stack competitive programming tracker integrating LeetCode, Codeforces, and CodeChef APIs, providing unified analytics dashboards, contest calendars, and an AI-powered coding assistant.
- Implemented OAuth authentication, badge/achievement system, performance analytics with charts, and a contest tracker — deployed on Vercel with CI/CD via GitHub Actions.

Smart Infant Monitor System | *Python, PyTorch, OpenPose, DenseNet-121, YOLOv11*

- Developed real-time computer vision system for SIDS prevention, detecting unsafe infant sleep postures with 85% accuracy using dual-pipeline architecture with OpenPose-Keras and DenseNet-121 classification.
- Implemented awake detection using YOLOv11 and Eye Aspect Ratio calculation; optimized training achieving 0.25 loss convergence and 85% validation accuracy with comprehensive data augmentation.

ANPR System | *Python, YOLOv8, EasyOCR, OpenCV* | [GitHub](#)

- Built end-to-end license plate recognition using YOLOv8 and EasyOCR, achieving 90% accuracy on Indian datasets with OpenCV preprocessing for varied lighting and angles.

Experience

Software Engineer Intern

Jan 2026 – Present

nanoFarm

- Redesigned and rebuilt company website from scratch using React, Tailwind CSS, and Node.js, delivering a fully responsive, modern UI with improved performance and user experience.
- Integrated OpenAI API to power AI-driven features within the platform, enabling intelligent query handling and context-aware responses for end users.
- Developed and deployed a plant disease detection model, training a CNN-based classifier and exposing it via a REST API endpoint consumed by the frontend.

AI & Data Analytics Intern

Jul 2025 – Aug 2025

AICTE-Shell-Edunet Skills4Future

- Developed and tested ML classification models (Decision Trees, Random Forest, Gradient Boosting) achieving 92% accuracy through rigorous data validation and quality checks on 500K+ row datasets.
- Created automated Python test scripts to validate data preprocessing pipelines and model outputs; built Power BI dashboards with end-to-end testing ensuring accurate real-time visualization.

Software Development & Operations Intern

Jun 2024 – Aug 2024

Nabin RCB, Birbhum, WB

- Built and tested a full-stack accounting system (Django, PostgreSQL) with unit/integration tests, cutting data entry time by 60% and boosting inventory turnover by 25%.
- Automated sales data validation using SQL and Python, improving operational efficiency by 15%.

Certifications

- IBM Generative AI Fundamentals Professional Certificate (Prompt Engineering, Foundation Models & Platforms, RAG, Ethics, Business Transformation) – Coursera/IBM (2025)
- CCNA: Introduction to Networks – Cisco (2025)
- Business Analysis & Process Management – Coursera (2025)

Leadership & Activities

- Executive & Associate, E-Cell Manipal – Coordinated 5+ technical workshops for 500+ participants (Sep 2022 – May 2024)